Experimental context limits transfer of microprotein sequence scores

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Abstract

We asked whether microprotein sequence scores validated for one experimental target remain useful after the assay, sequence input or decision changes. Using published data, we distinguish a core reporter benchmark from retrospective transfer extensions and exploratory diagnostics. Exact-sequence joins, dependence-controlled evaluation and source-only training define the comparisons. Among 1,072 development constructs, annotation-trained composition resemblance correlated negatively with reporter accumulation (Spearman $\rho = -0.292$), whereas outcome-trained composition and terminal-aware models reached 0.747 and 0.780. Tests using 1,044 identical peptides across reporter contexts, independent canonical tiles and 1,743 paired sequence changes showed that favorable within-context rankings did not imply reliable transfer. Terminal features could improve within-context prediction while reducing cross-context performance, and a model preferred by global correlation was not always preferred for top-decile selection. Source-population exclusions, matched inputs, training-exposure audits and strong published baselines also changed the eligible comparisons and narrowed apparent improvements. The practical implication is to validate a score against the intended measurement, input and decision, and to test paired effects before using a ranking model to prioritize edits. These results define specific transfer boundaries rather than a universal predictor. Native turnover and peptide-dependent activity remain separate experimental questions.

Keywords: microproteins; noncanonical open reading frames; protein stability; sequence predictors; reporter assays; cross-assay transfer; reproducibility

1 Introduction

Microprotein sequence scores can address three different questions: whether a sequence resembles an annotated protein, how much product accumulates in a reporter assay, and how a sequence edit changes the measured outcome. Performance on one question does not establish performance on the others.

Thousands of translated open reading frames fall outside conventional protein annotations. Standardized catalogues support their study [1, 2], but translation, peptide accumulation and peptide-dependent function require distinct evidence. Selected noncanonical peptides already have motif-dependent cellular phenotypes and native complex structures [3–5]. Conversely, an upstream ORF can regulate downstream expression through its RNA architecture or the act of translation [6]. Endogenous variant/pathway tests and engineered degron-exposure experiments provide further validation precedents [7, 8].

The six-feature XGBoost nORF Score showed no significant association with EggNOG Gene Ontology annotation [9]. Our controlled reanalysis retained compositional differences after exact length matching. We evaluate sequence representations against measured reporter outcomes and sequence interventions.

Annotation and detection predictors have their own intended targets. ShortStop prioritizes resemblance to curated microproteins relative to synthetic references [27]. PepScore integrates ORF features and was accompanied by individual expression assays [28]. Recent preprints distinguish Ribo-supported from MS-supported prediction or model repository MS detection using translation, conservation and peptide attributes [29, 30]. Reporter transfer evaluates a different endpoint; a faithful comparison must also retain any native RNA context and other inputs required by the original model.

Composition-based degradation prediction and sequence-context effects have established precedents. QCDPred predicts yeast quality-control degrons from composition, with companion work testing human reporters [10, 11]. Composition-only prediction also supports internal degron mapping [12]. Guo et al. relate composition, genetic-code and GC biases, disorder and terminal residues to reporter stability, including substitutions, extensions and paired reporters [13]. Cytosolic degron maps and terminal stability screens supply released PAP and DEGRONOPEDIA baselines [14, 18–20]. C-terminal sequence variation, composition-preserving shuffles and degradation signatures have also been studied [15, 16], with GC constraints as established context [17].

Calibration and richer sequence representations likewise require comparison with existing methods. Experiment-specific nonlinear calibration and variable feature gains are documented [21]. MAVE-NN and MoCHI model genotype–phenotype and measurement relationships, with limits on identifying their separate nonlinearities [22, 23]. Joint abundance–binding and abundance–activation models [24, 25], and experimentally tested multi-mutant improvements [26], further distinguish predictive fit from physical or functional interpretation.

The central question is when prediction validated in one context remains useful in another. We first establish the core annotation-versus-reporter comparison, then evaluate matched peptides across reporter contexts and independent canonical-tile and paired-edit targets. Strong frozen baselines, source-population rules and input/exposure controls bound those comparisons. Secondary diagnostics and exploratory candidate links are retained in S1 Text. The extensions are retrospective and outcome-informed; detailed methods follow the main findings.

2 Results

The core reporter benchmark, central transfer extensions and exploratory follow-up have different roles in the argument. All use existing data. The later extensions are retrospective and outcome-informed; their central narrative role does not make them preregistered or prospective confirmatory tests. Detailed results, sensitivities and candidate-source audits are retained in S1 Text.

2.1 Core benchmark: training target determines reporter prediction

On the 1,072 development constructs, the annotation-trained composition resemblance score correlated negatively with the GFP–microprotein reporter outcome ($\rho = -0.292$, 95% conditional interval -0.344 to -0.237). ShortStop's mean SAM probability had little monotonic association on its covered set ($\rho = -0.018$; $n = 1,057$). Models trained on the measured outcome performed substantially better: composition model A reached $\rho = 0.747$, and terminal-aware T reached 0.780 (Fig 1; Table 1). Useful composition prediction persisted in the strict additional library ($\rho = 0.719$). These results distinguish annotation resemblance from this reporter endpoint; they do not invalidate the original annotation task.

Adding native-order features to T gave a small primary rank increment, $\Delta\rho = 0.0040$ (95% conditional interval -0.0046 to 0.0115). The increment varied across contexts. Coverage, dependence groups, catalogue matching and all order sensitivities are reported in S1 Text and Figs S1–S3.

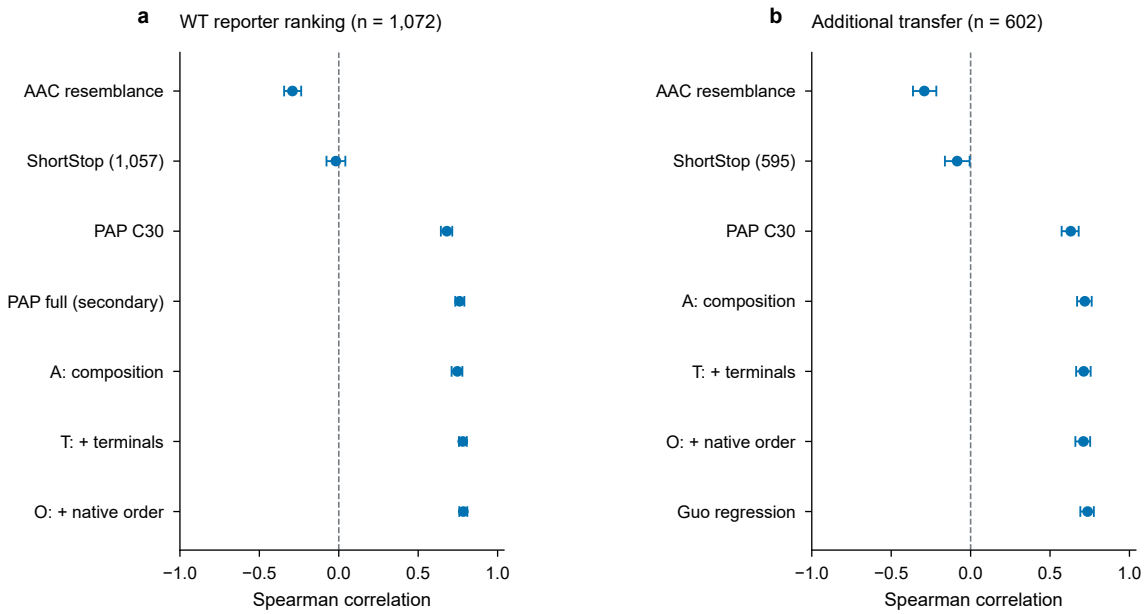


Fig 1. Reporter ranking and within-study transfer. Points show Spearman correlation; bars show 95% operational-group bootstrap intervals conditional on fitted predictions. Raw annotation and PAP scores are ranking comparators, not reporter probabilities. A/T/O are group-held-out in panel a and trained on development WTs for transfer in panel b. Coverage exceptions are labelled. The released Guo comparator is training-sequence/group-disjoint on these additional cargos, but not demonstrably prospectively blinded.

Table 1. Group-held-out native prediction and strict additional transfer. MAE is shown only for direct endpoint models on reporter scale. Additional sequences belong to the same publication, in a different library.

	GFP–MP WT, $n = 1,072$		Additional strict, $n = 602$	
Model	ρ	MAE	ρ	MAE
L: length	0.207	0.2261	0.167	0.1743
A: composition	0.747	0.1425	0.719	0.1259
T: plus terminals	0.780	0.1362	0.712	0.1311
O: plus native order	0.784	0.1354	0.710	0.1314
T plus shuffled order 1	0.770	0.1388	0.703	0.1344
T plus shuffled order 2	0.775	0.1380	—	—
T plus shuffled order 3	0.771	0.1378	—	—

2.2 Central retrospective transfer: the same peptides behave differently across contexts

Matching 1,044 peptide identities fixes sequence-cohort composition in the comparison of Guo reporter contexts [13]. Observed rank agreement was $\rho = 0.522$ between the original GFP–MP and MP–mCherry configurations, compared with 0.867 between the two GFP libraries (Fig 2). Source-trained T reached $\rho = 0.560$ on MP–mCherry, versus 0.742 for target-trained T; reverse transfer reached 0.566 on GFP–MP, versus 0.767 for target-trained T (Table 2). Composition alone transferred better than T between the original reporter configurations, reaching 0.663 and 0.648 in the two directions.

The added terminal information therefore improved some within-context rankings without uniformly improving transfer. Reporter identity, orientation, linker/tag architecture and library conditions co-vary, so this comparison does not isolate their causal contributions. Target-supervised calibration did not remove the target-scale error gaps and is adaptation, not independent validation of a frozen model.

Table 2. Terminal-aware T transfer on the same 1,044 peptides. G: original GFP–MP; M: MP–mCherry; U: mutagenesis-library GFP–MP. Rank transfer uses the source model directly. MAE transfer uses target-scale isotonic calibration learned entirely within outer training groups. The target reference is trained on that target in the same group folds.

	Target rank correlation		Target-scale MAE	
Transfer	Source-trained	Target-trained	Calibrated transfer	Target-trained
G to M	0.560	0.742	0.2062	0.1572
G to U	0.691	0.673	0.1211	0.1305
M to G	0.566	0.767	0.1820	0.1401
M to U	0.528	0.673	0.1492	0.1305
U to G	0.737	0.767	0.1453	0.1401
U to M	0.546	0.742	0.2065	0.1572

2.3 Central retrospective transfer: source and decision matter on independent cohorts

Composition models trained in the two nORF reporter configurations reached $\rho = 0.816$ and 0.817 on 14,143 eligible canonical tiles from 383 genes. The corresponding T correlations were 0.812 and 0.656 (Fig 3; Table 3). The GFP-source A-minus-T contrast was small and uncertain,

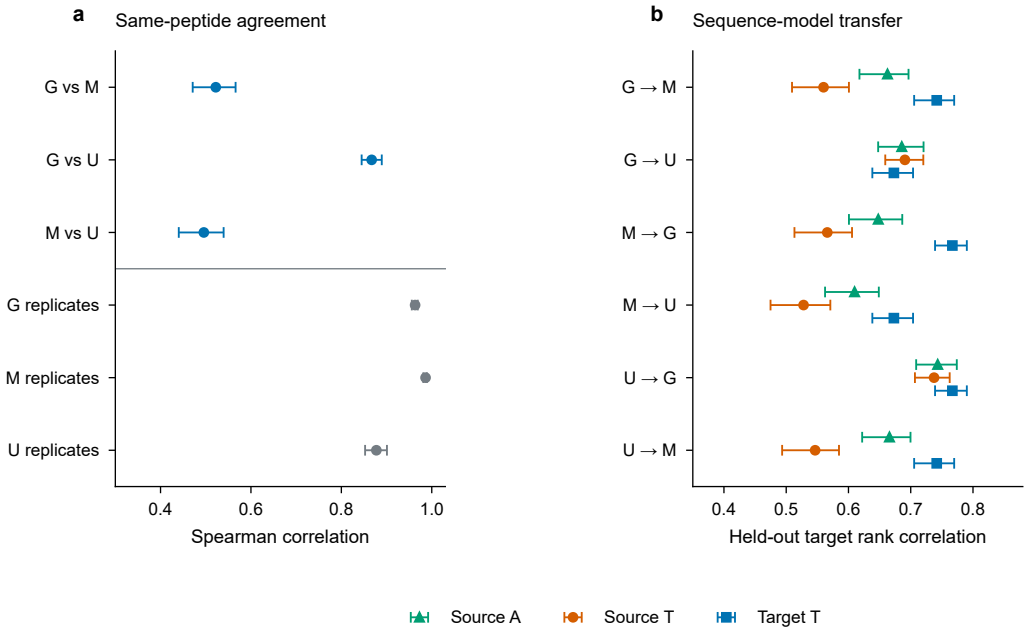


Fig 2. Same-peptide agreement and model transfer. a: correlations between context means and within-context biological replicates on the common 1,044-peptide population; missing replicate pairs are omitted as specified in the text. b: all six directed transfers of source-trained A and T versus target-trained T on identical held-out groups. Bars are descriptive 95% operational-group bootstrap intervals conditional on saved predictions. G: original GFP–MP; M: MP–mCherry; U: mutagenesis-library GFP–MP. This is an exploratory same-study comparison.

whereas the MP–mCherry-source gap was 0.161 (95% conditional interval 0.149 to 0.174). This supports source-specific transfer behavior, not a general claim that terminal features are harmful.

The decision metric also changed the comparison. GFP-trained T selected higher-scoring top-decile tiles despite its slightly lower overall correlation. Global rank association and prioritizing a subset are therefore different evaluation targets. Frozen normalized QCDPred reached $\rho = 0.800$, narrowing the composition-head advantage; the top-decile selected-mean difference intervals spanned zero.

Table 3. Source-only nORF models evaluated on independent canonical tiles. All rows use the same 14,143 held-out tiles from 383 gene groups. Selected means are measured target-assay scores among the highest-scoring 1,415 tiles under each frozen policy; they are not source-scale predictions.

Source training	Model	External ρ	Selected assay mean
GFP–MP	A	0.816	0.5976
GFP–MP	T	0.812	0.6121
MP–mCherry	A	0.817	0.5970
MP–mCherry	T	0.656	0.5506

Independent paired changes supplied a separate intervention test. On the 1,743 matched 58-mer pairs retained after common source/PAP exclusions, source-WT calibration improved linear full58 PAP from $\rho = 0.651$ to 0.692. Raw published nonlinear PAP already reached 0.729, and its additional calibration gain was 0.006 (95% conditional interval -0.002 to 0.014; Fig 4; Table 4). Thus the specified linear improvement does not establish a general calibration advance.

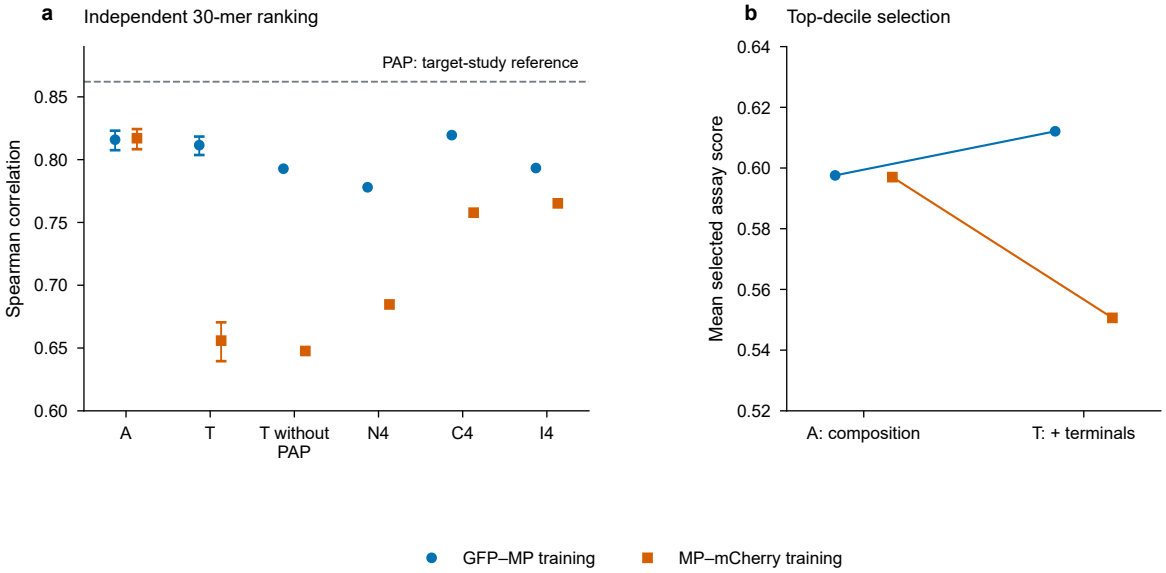


Fig 3. Source context and decision metric change the transfer comparison. a: independent-target Spearman correlation for both source-trained heads. Intervals for A/T are conditional 95% gene-bootstrap intervals; other diagnostics are point estimates. The dashed PAP reference was trained in the target study. N4/C4/I4 each add 80 positional columns to A; T without PAP removes its pretrained components. b: mean measured target score among each policy's highest-scoring 10%; policy contrasts and uncertainty are reported in the text and S9 Data. The source of the 30-mer assay is independent; the target objects are canonical tiles rather than native nORFs.

PAP's author-held-out canonical-tile performance is a target-study reference, not independent external validation of PAP.

Table 4. Independent paired-effect ranking for frozen baselines. All entries use the same 1,743 pairs. Calibration is learned solely from Guo GFP–MP WTs. Full58 and C30 inputs are distinct pre-specified representations. Both endpoints of every pair are 58 residues. Values are Spearman correlations with observed RT-minus-WT change; all components/sensitivities are in S10 Data.

Model and input	Raw	Source-WT calibrated
QCDPred normalized	0.602	0.622
Human25 normalized	0.662	0.655
Linear PAP, C30	0.480	0.510
Linear PAP, full58	0.651	0.692
PAP CNN1, C30	0.518	0.530
PAP CNN1, full58	0.729	0.735
PAP CNN7, C30	0.524	0.533
PAP CNN7, full58	0.725	0.725

2.4 Retrospective controls: population, inputs and exposure bound the comparisons

Eligibility rules changed an apparent feature advantage without changing the predictions. On all 1,014 valid Leu-to-Ala pairs, T exceeded A by $\Delta\rho = 0.097$. Applying the original terminal-Leu exclusion left 798 pairs and a contrast of -0.013 (95% conditional interval -0.032 to 0.005).

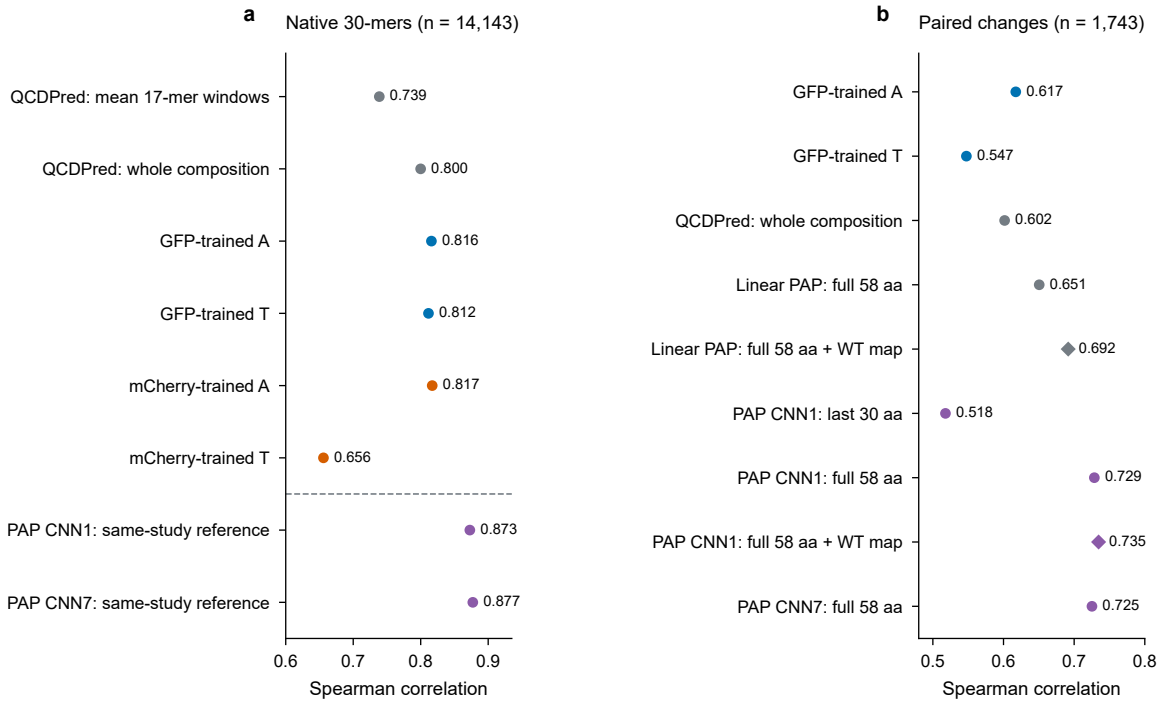


Fig 4. Published baseline challenge across two fixed cohorts. (a) Native ranking on 14,143 tiles/383 genes. The dashed separator distinguishes PAP neural models trained in that study from externally trained references. (b) Paired RT-minus-WT ranking on 1,743 independent 58-mer window pairs/1,591 groups after common source/PAP-training exclusions. Diamonds denote Guo-WT-only calibration. Full58 is supported variable-length inference beyond the training 30-mer length. Point estimates are shown; all variants, source contexts and conditional paired contrasts are retained in S10 Data.

The append-edit advantage persisted. These are post-hoc population contrasts using the source authors' rule [13], not a newly identified causal confound (Table S1; S1 Text).

Input coverage also mattered. On the independent pairs, raw CNN1 reached $\rho = 0.518$ on C30 and 0.729 on full58. The full58 representation extrapolates beyond the training length, and this comparison does not identify which residues or mechanisms explain the difference.

The terminal-model development-screen audit matched an endpoint in 1,454 of the 1,743 common-cohort pairs. Neither endpoint matched in 289 pairs. In that subset, DEGRONOPEDIA reached $\rho = 0.537$, compared with 0.521 for PAP on identical C23 inputs and 0.761 for full58 PAP (Fig 5; Table S2). The same-input difference interval spanned zero. The audit does not recover individual training-split membership or certify all historical exposure; its purpose is to bound the eligible comparison.

2.5 Exploratory challenges and candidate follow-up

The separate, selected FLAG-expression panel retained a failed transfer: primary MP–mCherry-trained T had $\rho = -0.652$ on the strict 20 constructs. Selection, construct differences and the detection endpoint prevent a population-wide failure estimate. All model/representation sensitivities, within-study edit analyses and conditional-order controls remain available in S1 Text and Figs S3–S6.

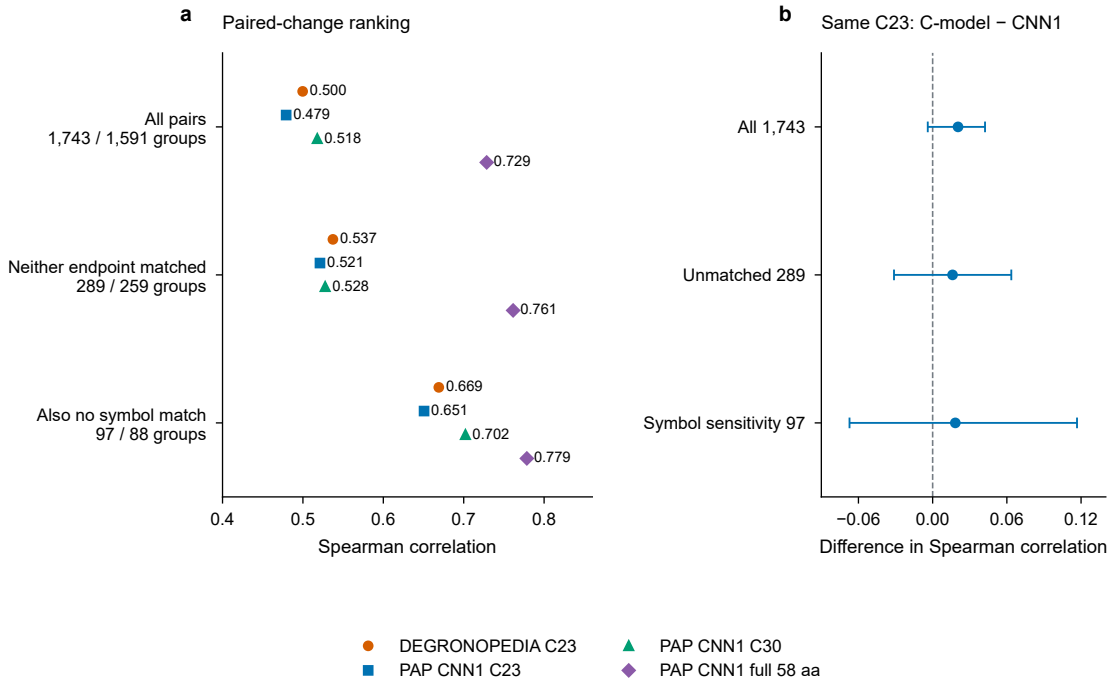


Fig 5. Predictor exposure and matched-input controls bound model comparisons. (a) Paired-effect rank correlations for the C23 terminal model and PAP CNN1 using C23, C30 and full58 on three nested populations. The original 1,743-pair cohort excluded source/PAP training exposure, while 1,454 pairs match an endpoint in the separate C23 development screen. Its actual individual 90/10 split is unavailable. (b) C-model-minus-CNN1 contrasts on identical C23 inputs, with 95% conditional group-bootstrap intervals. The matched-input controls are post hoc; all intervals span zero. The 97-pair available-symbol sensitivity is not alias-complete gene independence. Different cohorts do not identify an exposure effect, and C23/full58 neural inference extrapolates beyond the training length.

Exact sequence joins also nominated three follow-up candidates: RP11-620J15.3, JHDM1D-AS1 and the SLCO5A1 uORF peptide. Their existing Leu-to-Ala and append measurements increased reporter output in both source replicates, and the WT sequences have source interaction-tile annotations [13,33]. These exploratory links do not measure edited binding, native turnover or peptide-dependent activity. The full 60-candidate roster, source-positive/negative records, native-evidence reassessment and RNA-context qualifications are retained in S1 Text and S12–S18 Data.

3 Discussion

3.1 Transfer depends on the experimental question

The central result is that microprotein sequence scores transfer according to their training target, assay context and evaluation decision. Composition distinguishes the defined reference populations and predicts reporter accumulation when trained on that outcome. Annotation resemblance alone does not provide a portable accumulation score. Terminal features, calibration and residue order add predictive information in particular settings, without establishing a universally preferred representation.

The evaluation population can change the conclusion even when predictions remain fixed. The pooled terminal-feature advantage for Leu-to-Ala effects disappears under the original source exclusion, whereas the append contrast remains positive [13]. This limits generalization of the pooled result without identifying a causal molecular explanation. The intended decision also matters: GFP-trained T selects a higher-scoring top decile on canonical tiles despite its slightly lower overall correlation. Ranking unmodified peptides, selecting a subset and predicting an intervention are separate tests.

The matched-peptide analysis extends existing paired-reporter and same-sequence context comparisons [13, 41]. Fixing peptide identity removes a sequence-cohort explanation, but reporter identity, orientation and library conditions still co-vary; MP–mCherry also has a variable-length GS linker and an HA tag. The comparison therefore measures transfer behavior without isolating the responsible construct component. The selected FLAG-expression panel adds an adverse transfer case, but its selection, architecture and detection endpoint prevent a population-wide failure estimate [28].

3.2 Input coverage and provenance shape predictor comparisons

Strong published baselines materially narrow the calibration result. Source-WT mapping improves linear PAP change prediction, but raw nonlinear PAP performs better on the same independent pairs. The additional full58 neural-calibration interval spans zero, while the C30-to-full58 input change has a much larger association with performance. Full58 inference extrapolates beyond the training length, and this contrast does not identify a biological cause. These comparisons make model architecture and input coverage essential parts of evaluating a calibration claim.

Experiment-specific nonlinear readouts and separation of measurement from latent biological maps are established methods [21–23]. Physical interpretation requires additional anchors: Salinas and Ranganathan used affinity-characterized mutants and WT-affinity corrections, while their homologue comparison retained a hypothesized internal-equilibrium mechanism [42]. Joint folding/binding and abundance/activity models have their own physical assumptions and validation targets [24, 25]. Higher-order prediction and domain-extension or ligand maps also use distinct proteins, substitutions and validation settings [43–45]. Our source-WT remapping has no independent molecular anchor and does not recover an assay-independent physical stability scale.

Training exposure determines which model comparisons can be called independent. The terminal-model audit restricts the eligible pairs, and matched C23 inputs remove most of its apparent point-estimate gap against CNN1. Intervals spanning zero demonstrate neither equivalence nor a universal winner. Higher correlations after exclusions do not establish that exposure caused inflation. Population and decision-metric effects have precedents [26,46]; the added result is the specified shared-population comparison with inspectable provenance. Likewise, the small property-feature increment cannot be generalized to all order-aware models. CNN7 retains an order increment on its source study's held-out tiles, while independent paired-effect intervals span zero. Model shuffles test conditional predictive information, not the biological effects of physically shuffling a peptide.

3.3 Reporter prediction and native function require different evidence

Sequence descriptors alone do not identify a degradation mechanism. Composition-conditioned motif prioritization, degron scrambles and stabilized full-length products already have experimental precedents [11, 12, 15]. Terminal screens distinguish encoded sequence from exposed ends and pathway dependence from total stabilization [19, 20]. Recognition, ubiquitination and proteasomal engagement are distinct stages [47], potentially affected by accessibility and partner occupancy [8, 48]. Steady-state accumulation also depends on production: under a simple first-order balance, $P_{ss} = s/k$, where s is production and k includes degradation and dilution. Production and turnover measurements are therefore needed for a candidate-specific stability mechanism.

Annotation, detection and evolutionary origin address further distinct targets [17, 27, 29, 30, 49–51]. A peptide-only reporter test cannot refute a model requiring unavailable native RNA context. HLA presentation, intact-product accumulation and peptide-dependent function also require different evidence [35, 36]; missing detection or screen records are not established functional negatives.

The three candidate links define experiments rather than complete them. Synthetic-peptide interactions need not reproduce native stability, concentration or compartmentalization [33], and the edited products were not tested for binding or activity. Characterized MIEF1-MP70 and MRPL33_65 controls have native structural evidence [4], but neither has an exact primary GFP/edit pair in our bridge. Selected rescue and assembly studies establish functional non-canonical products [3, 5, 52–54], while abundance and functional readouts can diverge [55, 56]. JHDM1D-AS1/DHX15 and RP11-620J15.3 RNA studies do not isolate the nominated peptides [57–59]; the reported DHX15 turnover assay measures DHX15 rather than JHDM1D-AS1 peptide clearance. RNA architecture, translation and released-peptide activity require candidate-appropriate controls [6, 28, 60].

3.4 Practical implications

For reporter accumulation, choose or validate a score against the actual measurement and construct context; annotation resemblance or detection probability is not an interchangeable endpoint.

For cross-study comparisons, retain the input sequence seen by each model, audit training exposure and compare strong frozen baselines on a common eligible population. Report coverage and the intended decision metric alongside global correlation.

For edit prioritization, evaluate paired changes in the relevant sequence population. Increased reporter output does not establish slower native clearance or preserved activity; those require product-compatible production, turnover and functional measurements.

3.5 Limitations and next experiments

Several extensions were retrospective and outcome-informed. We use aggregate assay releases, operational dependence groups and intervals conditional on fitted models, rather than raw-data reconstruction, exhaustive remote-homology control or full retraining. Historical training snapshots are incomplete, and post-hoc intervals are not multiplicity-adjusted discovery tests. Canonical tiles independently test our source-trained heads, but their PAP-development provenance prevents treating the accompanying PAP scores as independent validation of PAP. The

three-row paper/release discrepancy is recorded. The common paired cohort audits source/PAP genes and windows, without certifying independence for every predictor. The terminal-model audit lacks individual split membership, does not resolve every alias or remote homologue, and does not audit N-terminal exposure. Comparisons must retain these coverage and provenance limits. The claim-focused literature assessment does not establish exhaustive priority.

Native validation should first identify the intact or processed product with an edit-compatible measurement and distinguish production from clearance. WT, the individual edits and the unmeasured combined edit/append panel should be tested separately. Binding or activity should be compared across overlapping measured protein amounts, with a predefined equivalence criterion for preservation. RNA, translation and recoded-peptide rescue controls should separate transcript, translation-process and peptide-sequence explanations. A masking hypothesis additionally requires a validated partner perturbation and a clearance measurement. These experiments could establish a candidate-specific biological relationship; the present work supplies their nomination and the measured prediction-transfer limits.

4 Conclusions

Microprotein sequence scores should be validated for the measurement and decision they will support. Composition transfers usefully in several reporter settings, while gains from terminal features, calibration and residue order depend on the source, input and endpoint. Strong published baselines, matched inputs and predictor-specific exposure controls are therefore necessary to interpret an apparent improvement. Exact candidate links nominate native experiments that test production, clearance, interaction and activity separately. The contribution is a reproducible empirical account of these transfer boundaries; native peptide-dependent function remains to be measured for the proposed edits.

5 Materials and methods

5.1 Analysis hierarchy

The core benchmark comprises the original dependence-controlled reporter evaluation. We distinguish it from the later same-peptide, canonical-tile and independent paired-effect transfer extensions. Those extensions are central to the present scientific question but retrospective and informed by earlier results. Source-population exclusions, input/exposure challenges, order/calibration sensitivities, the selected FLAG panel and candidate joins are explicitly secondary or exploratory. These labels describe analysis roles, not preregistration or prospective confirmatory status. All outcomes, including adverse transfer cases, remain reported in S1 Text.

5.2 Controlled catalogue context and clustering reproduction

The controlled catalogue analysis reconstructed 28,359 TransCODE Phase 2 candidates against hg38, audited current GENCODE v50 coding overlap and retained 27,670 eligible nORFs. Canonical representatives were the longest valid translated isoform per gene (19,563 genes). The protr descriptor definitions were versioned [31]. Default lag/lambda 30 restricted the all-family comparison to proteins of at least 31 residues; no missing long-lag features were imputed. Exact full-protein length matching after dependence filtering yielded 744 comprehensive pairs,

245 Primary pairs and 472 pairs with an additional absolute GC caliper of 0.02. Matched pairs and linked sequence/gene groups stayed in the same evaluation fold.

Clustering used the 1,488 comprehensive matched sequences, all 1,440 descriptors, equal-feature or equal-family weighting, PCA retaining 50 components, $k = 2, 3, 5, 8$, multiple seeds and 80% subsampling. Native sequences were compared with composition-preserving shuffles. A separate computational rerun checked saved PCA memberships and two t-SNE maps. This reproduces the controlled computation; the original cluster memberships remain unresolved. UMAP was not rerun. The original matrix and complete historical settings remain unavailable.

5.3 Experimental sources and target definitions

Guo supplementary tables supply normalized fusion-reporter scores [13]. The primary development set comprises 1,072 valid wild-type (WT) GFP–microprotein (GFP–MP) cargos; 1,071 have both reported biological replicates. The opposite MP–mCherry orientation supplies 1,046 measured WTs. A separate GFP–MP mutagenesis library contains 1,072 WT baselines, 730 HLA-associated and 71 digest-MS-associated natural sequences, and engineered variants. The additional 801 sequences are absent as exact peptides from the released 1,072-row Guo regression-training table. They remain within the same study, and their availability does not establish prospective blinding of the published models.

We used released means, available biological replicate cells, oligos and source-code metadata. The publisher's article-plus-supplement file subsequently provided full STAR Methods access [13]. The source describes within-bin counts-per-million normalization, five bin weights (0.1, 0.3, 0.5, 0.7, 0.9), their counts-weighted average and replicate-specific min–max scaling. We did not rerun that raw read processing or infer censoring from aggregate scores. Raw read-depth and censoring information were not obtained for this reanalysis. These limitations constrain error and detection interpretation. Throughout, "reporter score" denotes the supplied assay measurement, not an endogenous half-life, thermodynamic folding stability or function.

Exact sequence editing verified 1,014 global Leu-to-Ala pairs and 1,007 extensions by the constant C-terminal sequence DNYNEPKANQ. Observed change was mutant minus WT score in the **same mutagenesis library**. Original and mutagenesis-library WT measurements differ, so the former were not substituted as mutant baselines. Missing outcomes or predictors were not replaced by zero. These all-sequence-valid populations are broader than the source's specified Leu-to-Ala analysis: its STAR Methods exclude WTs containing Leu at either of the last two residues, and Fig. 4F states this restriction for the WT/substitution/extension comparison [13]. After recovering the full methods, we applied that WT-sequence rule to both edit families as a post-hoc sensitivity, retaining all earlier fits and predictions. This retained 798 Leu-to-Ala pairs in 611 operational groups and 802 append pairs in 623 groups. All, retained and omitted strata, mean and both replicate effects, all 20 original predictors and compatible-scale errors are supplied in S12 Data. T-minus-A, O-minus-T and calibrated-minus-raw full-PAP rank contrasts use 1,000 group resamples conditional on the original fits, with no multiplicity adjustment. An independent workbook reader checked all 2,021 membership flags; an independent rank implementation checked 360 point metrics and 18 point contrasts. Bootstrap distributions were not independently replayed. This changes the evaluation population, not the fitted models, and does not reproduce every original assay-processing decision.

The separate Yang assay contains 29 uORFs and eight other ORFs, with native lengths 17–129 residues [28]. We translated all 37 supplied synthesized constructs, verified their reported lengths

and terminal DYKDDDDK FLAG tags, and joined measurements by ORF name. Untreated outcomes are the mean of five replicates for the 29 uORFs and four for the other eight. The target is GFP-normalized FLAG-positive cell fraction, not the Guo numerical scale. The source describes the 29 uORFs as randomly selected across PepScores; the eight other ORFs were chosen with high scores. The combined 37-construct set mixes these selection schemes and ORF classes; it is not a population-wide validation sample.

5.4 Dependence audit and score provenance

Exact peptide identity, parental construct, available gene identities and catalogue overlap groups supplied dependence links. Engineered sequences shared by different parents connected those parents. We exhaustively aligned eligible pairs among native assay peptides, Yang natives, annotation-reference sequences and available ShortStop positive-reference counterparts. Local Smith–Waterman alignment used BLOSUM62, gap-open 10 and gap-extension 1. Primary edges required at least 60% identity over alignment columns and at least 80% coverage of **both** full sequences; a 40% identity graph was a declared sensitivity. Length and composition bounds only removed pairs that could not meet these criteria. Connected components defined split groups. These are operational dependence groups, not proof of no remote homology or pretraining exposure.

The annotation-trained AAC classifier retained the original fixed Extra Trees configuration: 200 trees, maximum depth 10, minimum leaf 10, square-root feature sampling, balanced classes and seed 100. Whole reference pairs and linked reference split groups were removed to a fixed point if connected to any Guo or Yang parent. Seeds 101 and 102 were sensitivity analyses. Higher scores indicate canonical resemblance, opposite to the nORF Score convention [9]; this classifier never fitted reporter outcomes.

ShortStop v1.2.0 (28 October 2026) precomputed scores were joined only after independent GTF translation against hg38 verified exact peptide identity. The primary continuous score was mean SAM probability across released models. Published standard and stringent consensus calls were retained as distinct threshold analyses. Actual supervised-positive identifiers were distinguished from ToBePredicted records, which the inspected training code excludes from supervision. The available 1,571 positive counterparts are current reviewed sequences, not a complete historical training-sequence archive. Thus absence of exact matches does not establish complete training or pretraining separation.

The frozen PAP human30 model was reimplemented from safely parsed published numeric coefficients [14]. Primary transfer uses the final $\min(30, L)$ residues. Composition terms scale residue counts to 30 positions; terminal, pair and triplet terms follow the released positional definition. The full-peptide normalized-composition version is a secondary transfer, not a newly fitted PAP model. Deposited Guo classification/regression predictions were evaluated only on the additional natural sequences and their strict subset, not against other methods' held-out predictions on Guo's own training rows. Supplied PepScore values were used only for the exact Yang constructs; missing transcript, domain or evolutionary covariates were not invented.

5.5 Endpoint models, order controls and nested validation

Models used five outer and three inner group folds. Groups were randomly tie-broken with seed 20261031, sorted by descending size and greedily assigned to the smallest fold without using

outcomes. Ridge penalties $\{0.1, 1, 10, 100, 1000\}$ were selected by inner mean absolute error (MAE). Standardization and intercept fitting used each training split only. All representations shared the same outer folds. Predictions and fit receipts were saved for every held-out row.

The length model L contains log length and its square. A adds all 20 AAC frequencies. T adds first-five and last-five residue one-hot encodings, whole/N10/C10 hydropathy and PAP composition/terminal components. O adds 40 short-range property-product averages: eight default Moreau–Broto AAindex properties at lags 1–5, standardized across amino acids, then centered and scaled against 199 within-peptide shuffles. Shuffles preserve exact composition and the first and last five residues. Zero null variance produces a zero residual. This avoids lag-30 imputation and supplies features for short cargos.

Three independent shuffled-order representations add the same 40 standardized features to T using additional shuffled realizations. They test representation effects conditional on composition and terminal residues; the shuffled peptides have no measured outcomes and are not labelled experimental negatives. Raw order features, two extra fold seeds, 40% identity grouping and fixed Extra Trees regression for A/T/O were declared sensitivities. The tree regressors used 200 trees, depth 10, leaf minimum 10 and feature fraction 0.7. No outer-fold winner was substituted for the primary Ridge analysis.

Primary contrasts are O-minus-T rank correlation and T-minus-O MAE, with 1,000 operational-group bootstrap resamples conditional on saved predictions. These intervals do not include full model-retraining uncertainty. Additional intervals and subgroup results are descriptive; no secondary significance claims or multiplicity-adjusted discovery list is made. No equivalence test was performed. Median absolute biological-replicate disagreement and half that value provide descriptive scale context, not a formal noise floor or biological relevance threshold.

5.6 Calibration, perturbation prediction and independent transfer

Frozen scores were mapped to reporter scale using isotonic regression fitted to training WTs within each group fold. Direction was determined by training correlation and recorded, including reversed AAC-resemblance maps. Raw score MAE was not compared across incompatible scales. For mutations, each parent's WT and variant were predicted with the **same parent-held-out** model or isotonic map trained on mutagenesis-library WTs. We report raw rank differences as well as calibrated prediction changes. Zero change and an out-of-group training-family mean are error baselines. Replicate-specific outcome changes were analyzed separately.

Secondary mutation-trained Ridge models use mutant-minus-WT changes in A/T/O features and grouped nested evaluation within each family. Opposite-family transfer trains on the other edit family after excluding all held-out parent groups. This distinguishes native-score extrapolation, learning an edit family, and transfer between edit families.

Before scoring Yang outcomes, main predictions and model choices were saved in a validation-release receipt. Primary direct transfer trained on MP–mCherry WTs, whose cargo has a free N terminus and masked C terminus, as an approximate context choice for a C-terminal FLAG construct. Actual FLAG-fusion sequence features were primary; native-only features and GFP–MP training were declared sensitivities. Every training group connected to any Yang construct was removed from these direct-model fits. We did not fit, invert, calibrate or select models on the Yang outcomes. The representation is still a cargo approximation: neither Guo carrier architecture is identical to a FLAG construct. Analyses report all 37, the 27 without exact Guo overlap,

and the 20 passing strict separation from all retrieved Guo natural cohorts. Rank correlation and AUROC at the authors' 5% expression threshold are reported; cross-assay absolute MAE is suppressed.

5.7 Exploratory paired reporter-context analysis

To compare contexts on the same peptide population, we joined WT records by both source identifier and exact native amino acid sequence across original GFP–MP (G), original MP–mCherry (M), and mutagenesis-library GFP–MP (U). The complete-mean intersection contains 1,044 peptides in 783 primary dependence groups. Of 1,072 G, 1,046 measured M and 1,072 U records, respectively 28, two and 28 fall outside this intersection. Biological-replicate agreement was computed on available replicate pairs within this same population. The extension was specified after inspecting aggregate paired correlations; it is exploratory and does not supply independent-study validation.

All six directed transfers used the previously defined A and T representations and nested Ridge settings, with identical five-fold assignments across the three contexts. This common-cohort split differs from the earlier 1,072-row development split. Every model excluded the test peptide's entire dependence group across contexts. The target-trained reference used target measurements from the training groups; the source-trained model used source measurements from those groups and was evaluated against the held-out target ranking. For target-scale error, isotonic calibration used source predictions generated by a further group cross-fit within each outer training partition, paired with training-target measurements. Its monotonic direction was learned there. Neither source fitting nor calibration used outer-test target outcomes.

A separate measured-source bridge fitted an isotonic map from paired training measurements and required the test peptide's measured source score; it is not a sequence-only method. Training-target means supplied an MAE baseline. We repeated the sequence analysis with the prior 40% identity graph and split seed 20261032. Descriptive intervals use 1,000 group-bootstrap resamples conditional on saved predictions, including paired contrasts on identical target rows. No multiplicity-adjusted discoveries are claimed.

5.8 Independent canonical-peptide transfer extension

After examining the paired-context results, we specified a new comparison before evaluating predictions against an independent target. We used the author-held-out cytosolic 30-mer library from Voutsinos et al. [14], pinning the repository at commit `9b3eafa6bd9730c558cd018063fa0180edf9db1a`. The released split and original processing log both contain 15,818 test members, whereas the paper states 15,815. We retained all valid released members before exclusions and report this discrepancy. Exact author-training peptide duplicates were absent. Excluding genes shared with the retrieved Guo/annotation inputs and peptide matches at $\geq$ 60% identity and $\geq$ 80% coverage of both sequences left 14,143 tiles from 383 ENSG gene groups. The 40% identity sensitivity retained 13,730 tiles. These operational rules do not guarantee absence of remote homology.

A and T heads were fitted separately to the original 1,072 GFP–MP and 1,046 MP–mCherry WTs, using the benchmark's source-group tuning and training-only scaling. All target predictions and input/model hashes were frozen before joining released abundance scores. Target outcomes did not select the source, sign, penalty or representation. A/T source features were

independently reconstructed and matched the saved benchmark exactly. We report Spearman correlation and paired A-minus-T contrasts, with 1,000 gene-cluster bootstrap resamples conditional on fitted predictions. A second source-only tuning seed is a sensitivity analysis; target-scale error was not compared across these different assays.

Secondary diagnostics remove PAP from T or add 80 one-hot columns to A for positions 2–5 (N4), the last four residues (C4), or the four central residues (I4, zero-based slice $L//2-2 : L//2+2$). Equal nominal dimension does not equalize effective complexity or isolate terminal exposure. Source grouped nested-CV results accompany these diagnostics. PAP internal/full scores are target-study-trained held-out references, not equally external competitors; their coefficients also contribute to T's pretrained features.

After initial transfer inspection, a secondary policy comparison reused the earlier benchmark's 10% selection fraction without target optimization. Each frozen score selected its highest-ranked $\lceil 0.1n \rceil$ tiles; ties followed lexical peptide order. We report mean measured assay score among selections and paired policy contrasts, recalculating selection in each gene-bootstrap draw. This is retrospective assay-rank prioritization, not prospective candidate or functional validation (S9 Data).

5.9 Frozen published baselines and independent paired changes

After the canonical-tile analysis, we added two secondary challenges with protocols and prediction hashes recorded before their new model–outcome associations. Earlier target results were already known. The original QCDPred code and labeled training tiles were pinned to the authors' repository commit `85979b1b4123b6b5391b617d16551969eda9f56e` [10]. We retained both the mean stability over complete overlapping 17-mers and a whole-cargo composition representation. The latter rescales the sum of residue coefficients by $17/L$ before the published logistic transform; it is a declared alternative representation, not the ordinary 17-mer-window output. No labeled training 17-mer exactly matched the tested windows. This check is not an exhaustive homology certificate. The human25 composition model was retained with first-25 and whole-cargo normalization; its native-tile results are target-study-trained references (S10 Data).

Chu Supplementary Data 1 contains 3,249 matched pairs [15]. Both measured cargos have 58 residues. The RT-derived cargo comprises a retained WT suffix followed by the readthrough sequence; 339 rows retain no WT residues. These are terminal-window replacements, not simple appends to the measured WT58. We required unique gene mapping, standard nonidentical 58/58 sequences, consistent reported sequence/length fields, and at least 60 reads at both endpoints in both replicates. Six Excel-date gene-name artifacts were resolved by transcript/sequence mapping; two inconsistent WT/shared-suffix rows were excluded rather than repaired from outcomes. RT-minus-WT replicate means were checked against all released mean/delta columns. Source gene and sequence exclusions used the preceding 60% identity/80% bilateral-coverage rule, with 40% sensitivity. The common primary cohort additionally excluded PAP-training-linked genes and exact PAP30/human25 training windows from either endpoint. This retained 1,743 pairs in 1,591 operational dependence groups joining gene identity and identical endpoint sequences. The stricter cohort contains 1,738 pairs; the source-head-only sensitivity contains 2,326. Current-transcript terminal matching is an additional sensitivity, not outcome-based membership.

We froze A/T predictions fitted in each original Guo context, linear PAP composition/terminal decompositions, QCDPred and human25. For every frozen baseline, an increasing isotonic map used only the original 1,072 Guo GFP–MP WT outcomes; calibrated changes subtract mapped endpoint scores. No Chu outcome fitted or selected a source, direction, model or calibration. We retained all 22 original raw/calibrated score variants, clipping and zero mapped effects. Absolute errors on incompatible raw score scales are not compared (S10 Data).

5.10 Published neural PAP challenge and conditional order controls

We then challenged these results with published default CNN1 and released kernel-seven CNN7 PAP weights from the pinned Voutsinos release [14]. Other models' outcomes were already inspected, making this a secondary stronger-baseline challenge. The internal channel applies a kernel-one or kernel-seven convolution, global averaging and nonlinear dense layers; a separate nonlinear channel encodes the final five residues. The two channel scores are added. Original model classes and numeric HDF5 weights were used, adapting the Keras backend import and legacy layer loading only. Independent NumPy forward passes verified the preserved graph and weights. CNN1's internal channel is a nonlinear function of composition; CNN7 can retain local order. Differences between separately fitted networks do not isolate order because both fitted channels may differ.

Both final-30-residue and full-cargo inputs were specified and retained. Full58 uses supported variable-length inference, extrapolates beyond the 30-mer training length and is not the ordinary default multi-window CLI aggregate. Every component received its own Guo-GFP-WT-only increasing isotonic map. No target model fitting or score inversion was performed. The authors trained PAP on abundance scores transformed from the sort-seq distribution to the flow-cytometry fluorescence distribution [14]. Here, raw PAP denotes the frozen published output before our additional Guo-WT remapping, rather than an absence of original readout processing. PAP results on its own study's held-out native tiles are labeled target-study-trained references, not equally external competitors.

For each input, two deterministic sets of 19 shuffles preserve composition and the first/final five residues (seeds 20261105 and 20261106). CNN1 native and shuffle-mean scores are analytically identical, checked on fixed samples; CNN7 shuffles were explicitly evaluated. Native-minus-shuffle rank contrasts test information in the prediction conditional on these controls, not experimental outcomes of shuffled peptides. All 40 neural raw/calibrated variants were frozen before new associations. Independent verification checked 62,962 model-input scores, 15,504 sampled shuffled scores, 20 threshold maps and all 340 native/paired metric rows. Float32 calibration outputs were checked against float64 interpolation at the appropriate numerical precision. Intervals use 1,000 paired operational- group bootstrap resamples conditional on fitted predictions, without retraining uncertainty or multiplicity-adjusted discovery. Additional full58-versus-C30 and neural-versus-calibrated-linear intervals are explicitly post hoc, after inspecting point estimates (S10 Data).

5.11 Terminal-model exposure audit and matched-input challenge

We next evaluated the released DEGRONOPEDIA C-terminal model [18]. This secondary retrospective challenge followed inspection of other predictors; its protocol preceded the new terminal-model associations (S11 Data). Source and all three released model weights were pinned to commit `2703570145c3c53cf276b705e1a6cc59083dfd12`. We used the original descriptor and

native CatBoost inference functions with all 2,997 named model features. The source's unused feature-selection regular expression was not applied. Current dependencies reproduced all 12 rounded author examples; the authors' unversioned environment prevents an exact historical-runtime certificate. Only the C-terminal model was applied to the new target.

We recovered the 22,564-row Koren C23 screen through its primary redistribution in Hickey et al. Supplementary Table 5A, whose notes attribute columns A–L to Koren Table S7A [19,32]. All rows contain standard 23-mers and numeric PSI. Exact final-23-residue matches were checked separately for both endpoints of every existing 1,743-pair common-cohort member. A match establishes development-screen exposure, not membership of the unavailable individual 90/10 training split. We retained all pairs, either-endpoint matches, neither-endpoint matches, and a further sensitivity excluding available exact gene-symbol matches. Unresolved aliases and 24 original Excel-date gene-name artifacts prevent certified gene independence. Existing pair eligibility and gene/identical-endpoint groups were unchanged.

We froze scores for 3,302 unique C23 inputs before evaluating terminal-model associations. No retraining, target calibration, direction inversion or target-driven model choice was performed. Higher predicted PSI retained its published stability direction. WT, RT and RT-minus-WT rank correlations, both replicate changes and all frozen comparator scores were retained on every exposure stratum. An independent workbook reader verified every join; 32 original-descriptor preparations reproduced frozen scores exactly.

After inspecting the first terminal-model results, we added explicitly post-hoc PAP CNN1/CNN7 controls on the same C23 inputs. The original C30 and full58 scores remained unchanged. C23 and full58 use supported variable-length inference outside the training 30-mer length; they are explicit input representations rather than the default multi-window CLI output. No new calibration or sign choice was made. Independent NumPy forward passes checked all 6,604 new model-input vectors. All 48 initial and 16 added metric rows were independently recalculated. Contrasts use 1,000 paired operational-group resamples conditional on frozen predictions; they exclude retraining uncertainty and multiplicity-adjusted discovery. Cohort exclusions change the evaluated population and cannot identify a causal effect of development-screen exposure.

5.12 Exploratory exact-sequence link to interaction candidates

We linked the 60 distinct full peptide sequences in Sandmann et al. Supplementary Table S2 to all retrieved natural Guo assay records and existing edit pairs [13, 33]. Only whitespace and one terminal stop were normalized. Candidate names and Ribo-seq identifiers were retained as annotations; neither authorized a join when full sequences differed. All 60 candidates, including unmatched records, were retained. Original source Supplementary Table S3 supplied synthetic binding tiles, domain/SLiM annotations, tagged-expression and phosphorylation records. Exact tile occurrences were mapped within each matched WT, and every available edit was annotated for residue changes inside those tiles. A source-associated motif or domain does not establish a causal residue or kinase–substrate relationship. An independent workbook reader checked all 60 candidates, 74 natural-record links, 32 measured edit pairs and 10 tile/edit annotations. This post-hoc descriptive link involved no new fitting, enrichment test, functional-negative labels or edited-binding measurement (S12 Data).

5.13 Candidate-specific native evidence and later MS reassessment

We compared all 60 selected candidates with the original human-heart lncORF and uORF tables by exact full-peptide sequence [34]. Reported in-vitro translation, shotgun peptides, SRM targets and positive technical runs were retained as separate source fields; missing coverage and targeted nondetection were not assigned functional labels. An independent reader checked 2,674 source cells and 89 positive SRM run records across the matched panel. For JHDM1D-AS1, the public RefSeq transcript sequence was translated independently to check compatibility with the selected peptide; this did not verify historical plasmid sequence or full isoform identity (S13 Data).

We then matched candidates exactly to full sequences in Deutsch et al. Supplementary Table 9 before using their ORF identifiers to retrieve Tables 3, 6, 7 and 12 [35]. Source-reported final tiers, peptide sharing and missing entries were retained without new tier assignments. For the JHDM1D-AS1 shotgun claim, original ORF, peptide and Universal Spectrum Identifier (USI) linked the community benchmark's full PSM table to its curator-rating table [36]. Its Fig 1c label correction was checked [37]; no study-level plot was used to assign candidate confidence. An independent pandas replay checked all 60 candidates and 4,143 cells from the later tables. These are published-table audits, not new spectrum interpretation, raw-data reprocessing or independent experimental replication (S14 Data). For a further native-context comparison, all 60 candidates were matched by exact full sequence to brain-atlas Tables S1, S2, S3 and S5 [38]. An independent worksheet-XML reader checked all retained source rows and 407 matched cells, ignoring erroneous declared sheet dimensions without editing the workbook. No score association or new function label was assigned (S15 Data). Additional literature annotations preserve source-specific growth-screen flags and exact product identities from the Shepelev preprint and Schlesinger et al. [6, 39]. All 60 frozen sequences were also compared with the encoded polymer sequences in PDB 5OOL [4]; an independent reader checked the sequence joins and modeled residues. Partner-perturbation enrichment, native structure, peptide detection and cellular activity remain separate evidence fields. S17 Data records these annotations, selected original-source checks and the explicit limits of the scientific reading. Literature validation examples were separated by construct, measurement and intervention rather than pooled as functional labels. S18 Data records primary-source scope for evolutionary age, engineered terminal exposure and endogenous variant readouts. These annotations introduce no new candidate phenotypes or predictor outcomes.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

Data and Code Availability

Public assay data and predictor releases are cited in the references and S1 Text. A separate public reproducibility bundle is prepared with original study scripts, frozen derived outcomes and predictions, split/dependence records, publication artwork, source provenance and an offline point-metric verifier. It omits original source papers/workbooks, third-party implementation copies, model weights and runtime libraries; these must be obtained from the pinned upstream sources under their respective terms. The public bundle is therefore distinct from the full local

S1–S18 review collection described in S1 Text. The verifier does not refit models, recompute bootstrap intervals or reconstruct raw inputs. The corresponding author will insert the durable repository/release reference after publication succeeds; no accession or reuse licence is asserted here.

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Research scope

No new participant, animal or laboratory experiments were performed.

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Supporting information

S1 Text contains the extended benchmark specifications, complete detailed results, exploratory diagnostics and candidate-source audits, six supporting figures, two supporting tables and the S1–S18 Data guide. All adverse outcomes and sensitivity analyses remain available there.